

1 ENTERPRISE RESOURCE PLANNING

Enterprise Resource Planning software will be explained in this chapter. Attention is given to the advantages and disadvantages of ERP software. Enterprise Resource Planning can be considered as follow-up development coming out of Material Resource Planning.

The functions of Enterprise Resource Planning are presented in a diagram and are explained in the text. The flow of goods and the transformations they may undergo are explained in paragraph 1.4. In paragraph 1.6 the flow of goods is linked with Enterprise Resource Planning.

1.1 ENTERPRISE RESOURCE PLANNING SOFTWARE

Enterprise Resource Planning software (ERP software) can be defined as software supporting the whole range of operations by processing the information associated with goods flows and financial flows.

ERP software above all has an important added value since it facilitates integrated recording of information throughout the organization. The information kept on record by an organization is similar for most companies and can easily be accommodated within a standard package.

In the past, each department installed its own computer system. The personnel department used a personnel management system, the financial administration an accounting package, while several planning and warehouse systems were in use in the production department. The separate systems often worked to the complete satisfaction of departmental users, but the links between these systems led to many problems, high cost levels and poor flexibility. An integrated ERP package offers a solution to this problem.

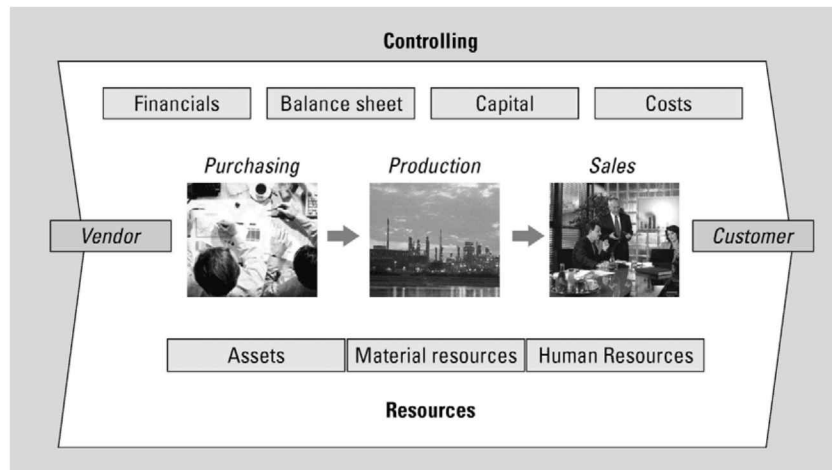


Figure 1.1 Enterprise Resource Planning in SAP S/4HANA

A large production company will be able in general to get by with a system for office automation (e.g. MSOffice), an ERP package (for example SAP S/4HANA) and some highly specialized software in the field of process automation (for example CAD/CAM systems).

The development of ERP software is a booming business. The wealth of advertisements and job recruitment sites like www.monsterboard.nl in publications specializing in computer and information specialist recruitment indicates a great demand for SAP consultants, Oracle experts, etc.

SAP (SAP S/4HANA, SAP ERP and SAP R/3) was in 2018 with a market share of 22% still the market leader. Oracle, who took over Peoplesoft and JD Edwards, is number two.

ERP systems attempt to offer a fully integrated solution for the management of financial flows, goods flows and information flows within organizations. Many ERP systems have their roots in production planning. This is sometimes apparent from the fact that the production module is significantly better developed than the other modules. The original nature of the production (discrete vs. process industry) often has a distinct impact over contemporary software. SAP originally started as software for finance and controlling and has consequently developed functionality features centered on production, logistics and HR. Nearly all the large players in the market are converging with respect to functionality of the ERP package. Meanwhile SAP has process modules designed to support the process industry.

1.2 ENTERPRISE RESOURCE PLANNING AS A FOLLOW-UP TO MRP

ERP is also seen as a supplement to MRP II. In the Material Requirements Planning concept (MRP I) a demand forecast is drawn up on the basis of bills of materials, calculated according to range of production orders and purchasing orders expected. Manufacturing Resource Planning (MRP II) was a later extension. This concept represents an expansion of MRP I into a more integrated planning and management concept. MRP II for instance also takes into account the available production capacities. MRP I and MRP II are compatible for use within a single organization. Increasingly far-reaching internationalization means that management must take place 'over the heads of individual plants'. Mutual deliveries between organizational units are included in the concept, for instance. ERP is compatible with this. The functionality of ERP software has been drastically expanded over recent years to include Customer Relation Management (CRM), sales support, Supply Chain Management (SCM), workflow management (WFM), Human Resources Management (HRM), etc. This trend continues; ERP software is further expanding its borders. Nearly all operational functions are now supported.

1.3 ADVANTAGES AND DISADVANTAGES OF EPR

Some of the advantages have already been discussed above. The most important of these are:

- *Integration*: the coherence between processes is seamlessly supported. Island automation is no longer an issue. Integration reduces problems caused by redundancy and errors in the data.
- *Process orientation*: instead of supporting activities in departments, ERP orientates itself towards operational processes that take place throughout the whole of the organization (and outside it), and indeed you see this in so-called Workflow Management solutions that the ERP systems offer. These cater to the support and direction of operational processes.
- *Openness*: ERP systems can communicate with other systems; this means that the Supply Chain concept, a far-reaching chain integration into the industrial process, can be utilized. ERP systems can also communicate with other systems, such as Customer Relationship Management software (CRM).
- *State-of-the-art*: the research being carried out by ERP suppliers is sufficiently comprehensive to ensure that a company using an ERP system can be confident that the functionality will both continue to be available and continue to be expanded. Associations of users often have some influence on aspects of functionality earmarked for expansion or improvement.

- *Industrial sector orientation:* ERP suppliers ensure that their software supports various sectors of industry. This support (called Industry Solutions) is constantly being expanded.
- *Internationalization:* ERP can be used supra-locational, supra-organizational and beyond national borders. This is an ideal solution for multinationals, since data taken from all their subsidiaries needs to be consolidated.

ERP is not a universal panacea. Naturally there are some disadvantages to the introduction of a system such as this. You might like to consider the following, for instance:

- *High costs:* the price of the software license is fairly high. The consultancy costs involved in the implementation of an ERP package exceed the license cost by a factor of approximately 2–4. A Dutch theme park, for instance, recently implemented an ERP system at a cost of about €3.5 million, of which €0.5 million was for the software license.
- *Lengthy implementation:* the time taken for making enhancements, customizing and the implementation often exceeds a year. Accelerated implementation techniques, Accelerated SAP, are available however, under which a selection is made of best practices in the implementation, where the need for labour-intensive fine tuning of the software is partly eliminated. In this case processes in the organization (as regards its operational processes) are adapted to the software. A solution like this is welcomed particularly by smaller organizations. In the case of medium and small companies, an implementation of this kind takes something in the order of 3 to 6 months, depending on the complexity involved.
- *No tailor-made solution:* ERP remains a standard solution: the user may have to modify his expectations as far as his package of requirements is concerned. Tailor-made systems are generally much more expensive than standard software. Modifications made to standard software are costly and can cause problems with upgrades. For that reason it becomes the policy of some companies to avoid modifications during the implementation.

Since the 1990s we have seen more organizations transferring to standard software. On the one hand, each company is individually aware of its own unique operational processes, yet these unique points are often remnants from the past. There is something to be said for taking a look to see whether the best practice options, which ERP systems offer, are not in fact better than one's own solutions. Why should my organization differ from the many other companies in my sector that have already been studied? Moreover, are my (unique) deviations from the general pattern as regards operational processes really as necessary as I think they are?

The employee learning period for new functionality in an ERP package is, as a matter of fact, generally much shorter than in cases where a totally new software package is acquired.

1.4 FUNCTIONS OF ENTERPRISE RESOURCE PLANNING

Figure 1.2 shows the most important functions that can be used by an ERP system in corporate operations. In this overview, the supporting functions are grouped around the logistics, the primary processes. The central feature is the flow of goods from the supplier to the customer: procurement results in the receipt and storage of raw materials (triangle), the various raw materials are processed during production into finished product units; these are subsequently delivered to the customer from stock. The way in which content is given to the primary processes of procurement, sales, stock control, shop floor control and manufacturing, is market and product specific. The flow of goods from the supplier to the customer is counterbalanced by a cash flow in the opposite direction: from the customer to the supplier. Recording of this cash flow takes place in the accounts receivable (AR) and accounts payable (AP) sub-administrations and in the ledger administration (general ledger).

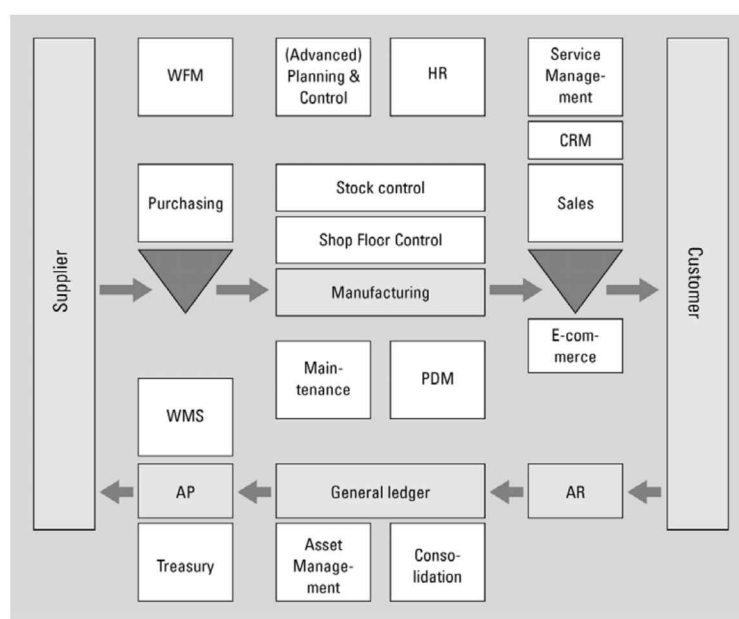


Figure 1.2 Enterprise Resource Planning

The relationship between the flow of goods and the cash flow constitutes the value cycle of an organization. The non-shaded functions form the secondary, supporting activities. Treasury, asset management and consolidation support the management and planning of financial resources over the short and long term. Workflow management, planning and control, human resources, service management and customer relations management support the logistics processes.

In Figure 1.3 the central focus is on the flow of goods. The planning levels and logistic modules are shown.

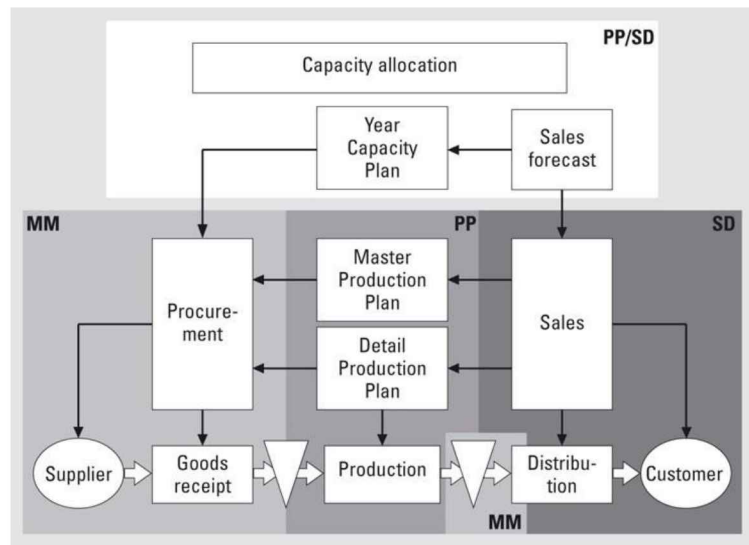


Figure 1.3 Integrated planning

In sales and distribution (SD) the sales statistics for the previous year form the basis for establishing a sales forecast for the coming year. This sales forecast makes up the demand for production capacity. By gearing the availability of capacity to sales demand, provision is made for the necessary capacity within the production planning (PP) by establishing both the nominative capacity and the expected annual capacity (year capacity plan and capacity allocation). The planning of the annual capacity constitutes the basis for establishing the main production plan within PP, as well as the basis for setting up the necessary procurement within Materials Management (MM). The master production plan has a planning horizon of one year and consists of production objectives relating to product units per period. The master production plan thereby creates a procurement requirement for raw materials/components. This requirement is met by establishing a procurement plan. The master production plan is converted into a detailed production plan, with a planning horizon of less than one year. The planning quantities are the number of units to be produced and the number of production orders. Even at this level, sales figures are coordinated according to production objectives and procurement. The planned production orders are converted into actual production orders, raw materials/components are issued and processed into finished products. The finished products are made available for delivery to the customer by MM. The planning for distribution to customers, which is also factored into this, takes place from within sales and distribution, SD.

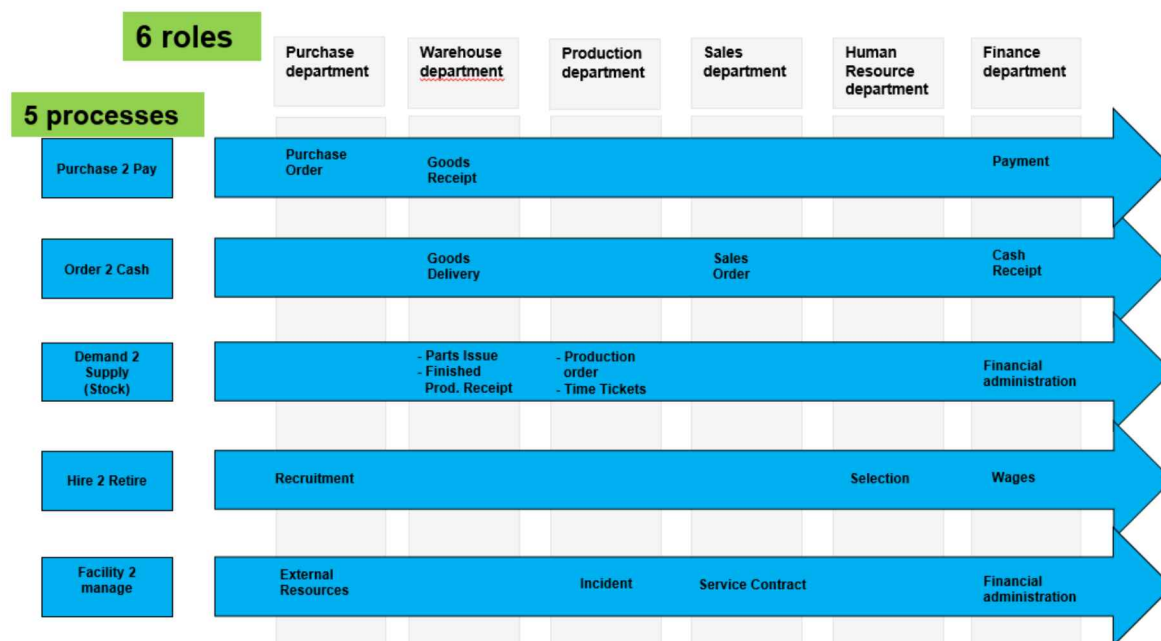


Figure 1.4 Integrated processes

Integrated planning comes with integrated processes cross departments as depicted in figure 1.4. In SAP you can define roles to organize the authorization and to assign groups of applications in the launchpad, part of the SAP user interface.

ment	Financial Accounting	Human Capital Management	Master Group	Materials Management	Navigation	Prodi
Curricula Material PDF Download Curricula	Display Financial Statement	Display G/L Account Balances	Manage G/L Account Master Data	Display Customer Balances	Maintain Business Partner	
						
Post General Journal Entries	Manage Journal Entries	Create Incoming Invoices	Display Supplier Balances	Post Incoming Payments	Post Outgoing Payments	
						
Reset Cleared Items	Assignment of Open Items	Post Parked Document	Manage Debit Memo Requests	Create Outgoing Invoices	Learning Snack FI Single-Choice	
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Figure 1.5 Role Financial accounting in the SAP Launchpad

1.5 SUPPLY CHAIN MANAGEMENT

The *Supply Chain concept* is central in SAP S/4HANA: the chain from the original supplier to the final consumer has to be managed. By bringing optimum co-ordination to the flow of information, the flow of goods and the cash flow, a maximum customer response is achieved at minimum cost according to SAP. It is possible for a customer, to gain information from the Internet regarding the products of a particular company, to adapt your own products accordingly and to place an order. This order then automatically ends up in SAP S/4HANA; where necessary, this package automatically generates orders to suppliers (multi-company planning) and even initiates the necessary processes within the organization of a company. In this way, open systems can be linked to each other and transactions extended over various individual companies.

The *Supply Chain* comprises all activities that have to do with the flow of goods and the transformations they may undergo, all the way from raw materials through to the end user, complete with the associated information flows (see Figure 1.6).

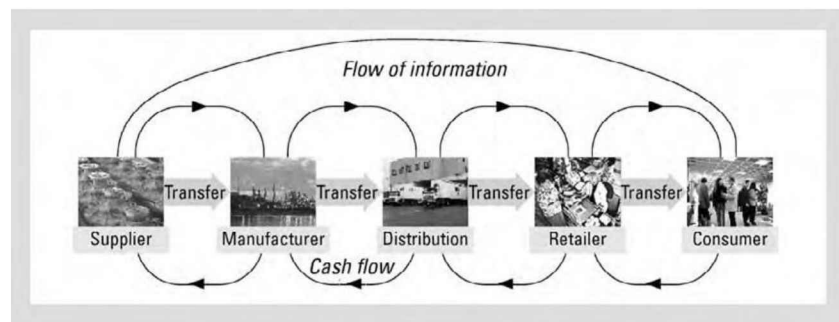


Figure 1.6 Chain from supplier to consumer

We are concerned here in fact with the activities that take place within the manufacturing process. In the case of simple products, this supply chain can readily be identified and surveyed, but when you think of the automobile industry, for instance, the situation is more complex.

Supply Chain Management (SCM) is therefore the control and integration of these activities by making use of all kinds of cross connections and networks in order to achieve a significant competitive advantage. The chain goes *upstream* in a *supplier network* and *downstream* in a *distribution network* (customers). The supplier network consists of all the companies that provide an input in any way whatsoever; these inputs could be physical raw materials, but they could also be information, for instance. In the case of the automobile industry, this chain would represent a supplier network of thousands of companies supplying goods ranging from steel and plastics, but also including complete gearboxes, brake systems, car radios and similar. Even the production of these gearboxes etc. is part of the supply chain.

Some authors say that when looking for the start of a supply chain you always come back to Mother Earth. The main point, however, is that each supply chain consists of a (generally large) series of linked suppliers and customers; each customer becomes a supplier in the next link of the supply chain until the product reaches the end user.

Seen from within an individual company, you can say that the three SCM perspectives are:

1. internal functions (transforming inputs into outputs);
2. upstream supplier functions;
3. downstream customer functions.

The *management challenge* is that coordination takes place between these three aspects in such a way that the company maintains its right to exist within the supply chain.

1. Internal functions

Where the internal functions are concerned, these relate to various operational processes, including the transformation processes of raw materials or semi-manufactures into finished products. These raw materials or semi-manufactures are supplied by the suppliers and the finished products are supplied to the distribution channel (the dealer organization in the case of the automobile industry). Orders from purchasers of automobiles must be translated into production orders. It is easy to imagine for instance that in the case of large freight lorries (as with DAF), this takes place on a one-to-one basis. Each customer order = one production order = one freight lorry. Even with private vehicles, things are beginning to move in the direction of production on demand. On deciding to purchase a new car, for instance, you can now specify the version, the color and whether it should have air-conditioning, etc. The influence of the final consumer over the production is steadily increasing.

The sales order data are very detailed: quotation prices, latest possible delivery dates, delivery scenarios, etc. Production planning is also extremely important. Use can be made of MRP here.

2. Upstream supplier functions

This refers to functions that take place outside the individual organization. Upstream is the procurement function. You think here of the selection of suppliers who can meet our requirements and wishes and who are reliable when it comes to delivery. The maintenance of good relations with suppliers is extremely important. In SCM, you see contacts with suppliers fanning out into the product development, commercial office, warehouse and administration departments and so on.

In this situation, there is close contact with the suppliers from within the internal operational processes.

3. **Downstream customer functions**

These functions comprise all distribution channels and processes used in facilitating the flow of products to the final customer. In most cases, it involves matters such as storage, transport, and sales activities.

SCM is orientated towards the control between various components in the supply chain. Five levels of integration are identified within SCM:

1. standardization of master financial data;
2. standardization of master logistics data;
3. exchange of time-dependent data;
4. standardization of planning and control procedures;
5. supply chain planning.

The fifth level can only be reached after the preceding levels have been completed.

Supply Chain Management can (and also will) lead to a curtailment of the number of links in the industrial column. This is clarified in figure 1.7.

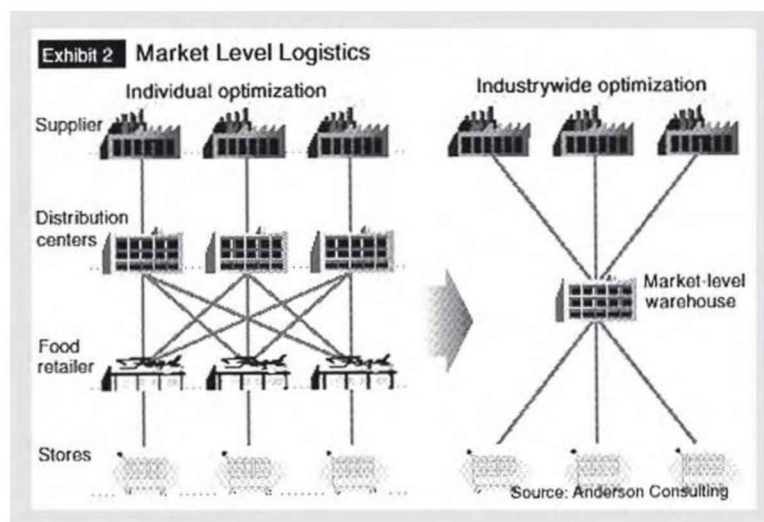


Figure 1.7 Supply chain planning

It is naturally the business of a particular company to ensure that it is not pushed out of the supply chain.

1.6 SUPPLY CHAIN MANAGEMENT AND ENTERPRISE RESOURCE PLANNING

Supply Chain Management has primarily to do with the building up and maintenance of close ties with suppliers and customers. You see that this is made possible by developments in ICT. SCM is therefore both *relationship-based* and *technology-driven*, to use a few words of ‘newspeak’. A combination of ERP and Internet (E-Business/E-Commerce) provides many options for the successful use of SCM. The advent of these technologies even makes a worldwide supply chain possible. As an example you can take the *DOW Chemical Company*, an industrial undertaking with branches in 168 countries and an annual turnover of more than \$50 billion in something like 3,100 products in the chemicals, plastics and agricultural sectors. The company employs 45,000 people and has 150 production locations in 35 countries. The most important aspect of SCM for DOW is that the company tries to offer consumers what they want at an acceptable cost (an acceptable price). The company has a *Supply Chain Technology Center* (based at Terneuzen in the Netherlands and at other locations worldwide) that will be responsible for cross-pollination between best practices and new technologies. The Center also scans for best practices and new technologies which DOW can also make use of. The deployment of ERP and a strong focus on operational processes is an important outcome of the Center’s activities.

Modern ERP systems integrate the planning, control and monitoring of processes within an organization. Because the internal flow of information between important departments within a company is integrally coordinated by ERP, it is possible to bring products to the market quicker, reduce costs and guarantee the loyalty of customers and suppliers. Using the right ERP system, a company can expand its ability to monitor its activities to the whole of the supply chain. The initiative of SAP in relation to Supply Chain Management (SAP SCM), for instance, covers all aspects of supply chain integration.

SCM solutions are on hand within SAP in the form of APO, Advanced Planner and Optimizer, and LES, Logistics Execution System. APO is responsible for improved demand forecasting and increased efficiency in production processes. LES facilitates more efficient, rapid and accurate flow of goods in the supply chain.

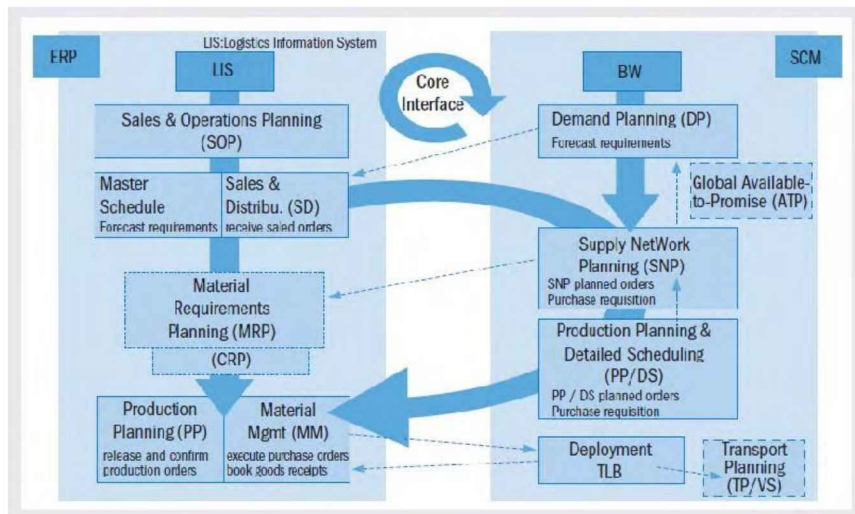


Figure 1.8 Supply chain planning

2 THE COMPANY: SAP AG

In this chapter the company that developed SAP S/4HANA and SAP is introduced and the products made by SAP AG are mentioned.

2.1 THE COMPANY: SAP AG

SAP AG is market leader in the area of inter-enterprise solutions whereby the internal processes and processes between companies and their vendors, customers and partners integrate.

The revenue of SAP AG was more than €27,55 billion in 2019. SAP employed more than 98,000 people in 2019.

SAP stands for Systems, Applications and Products. Some of SAP's clients are Bayer, Honda, Audi, KLM Air France, Nokia, Nestlé, Deutsche Bank, Siemens, Heineken, Quaker, Shell, Microsoft, Akzo Nobel, Rabobank, DSM, Mars and Dijklander Ziekenhuis.

2.2 PRODUCTS OF SAP AG

SAP S/4HANA is the successor of SAP R/3 and is an important step forwards, because this is the first service-oriented business application based on SAP Net Weaver. SAP S/4HANA can be used on-premise (own installation) or cloud (installation and maintenance by SAP).

SAP HANA (High-Performance Analytic Appliance) software is based on a in-memory database, is able to process huge amounts of data in real-time in the central memory. Complex queries on over 450 billion records can be performed in seconds. The processing of these 450 billion records used less than 3 Terabyte of physical memory. This database will be placed under SAP S/4HANA. Reports can be up to 9,000 times faster with HANA than with a conventional relational database.

SAP Fiori and Screen Personas to personalize and simplify the user experience (UX) for your SAP applications. Using modern UX design principles, SAP Fiori delivers a role-based, consumer-grade user experience across all lines of business, tasks, and devices. Use SAP Fiori apps, or take advantage of solutions that natively incorporate the UX – such as SAP S/4HANA and SAP C/4HANA (CRM systems).

SAP S/4HANA is the next generation ERP. The database HANA and the interface will be Fiori.

Midsized companies will be using the SAP S/4 or Business ByDesign.

For companies, which demand an extended functionality in a flexible and scalable application, these applications are an excellent solution.

Business ByDesign is a cloud solution from SAP, a complete on demand ERP system for midsized companies. The customer pays a fixed fee per user.

For smaller companies the *SAP Business One* application has been developed.

For customers, who want a simple but strong application which supplies support of their most important processes, Business One is the solution. This relative cheap, fully integrated software gives companies access to the information, application and services which are needed to enlarge their revenues, to control the costs and to stimulate their growth. SAP Business One integrates easily with other systems such as SAP Business Suite.

There is more information about SAP on www.sap.com. In the Netherlands there is an active users association, the VNSG (Dutch SAP user group). There is more information about the VNSG on www.vnsg.nl.

3 SAP S/4HANA

In this chapter the structure of SAP S/4HANA is explained and extensive attention is paid to the various solutions of the SAP S/4HANA system.

3.1 STRUCTURE OF SAP S/4HANA

SAP S/4HANA delivers functionality for business analytics, financials, human capital management, business performance and business services. Above that it gives support on the area of system management such as users maintenance, configuration maintenance and management of user experience and web-services.

SAP S/4HANA includes four individual solutions that support key functional areas, which together form the foundation for the business processes:

- SAP S/4HANA Financials (Finance and Controlling)
- SAP Human Capital Management (HCM)
- SAP S/4HANA Operations:
 - Procurement and Logistics Execution (Material Management, Warehouse Management)
 - Projects and Manufacturing (Project System, Production Planning)
- SAP S/4HANA Corporate Services:
 - Sales and Service (Sales and Distribution, Corporate Services)
 - Corporate Services (Enterprise Asset Management)

The different possibilities of SAP will be explained with the help of the Solution Map in figure 3.1.

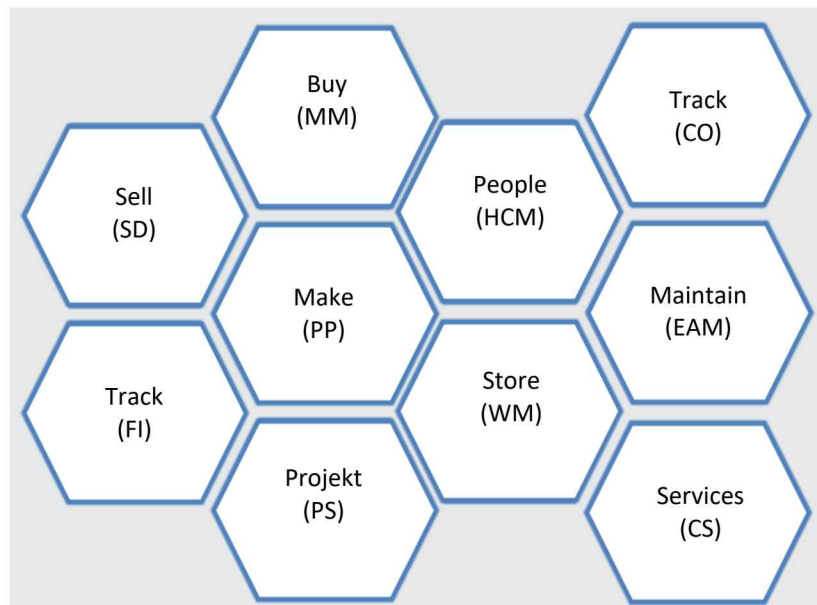


Figure 3.1 Solution Map

3.2 ANALYTICS

SAP Analytics Cloud (SAC) will be the analytic tool for all SAP systems. It can connect to HANA databases. Dashboards and predictive scenarios can be created with SAC. Dashboard are used for commercial and financial graphic overviews. An other good example of predictive analytics can be found in Enterprise Asset Management (EAM), companies do not like to do unnecessary maintenance and on the other hand do not like unexpected downtime of machines.

Sample: Summary Q1 (2014) – Q4 (2016)

This page outlines sales for the fictional company "The Best Run Juice Company". Feel free to click around and explore the data!

**Figure 3.2 Analytics**

3.3 FINANCIAL

SAP S/4HANA Financials provides a complete financial management solution for a broad range of industries. The solution offers depth of functionality in accounting, reporting, analysis, financial supply chain, corporate governance and treasury management. In the new concept within SAP S/4HANA is profit center accounting, determination of the costs of goods sold, special valuation accounts and consolidation accounts integrated. This integration supports the need for the multi-dimensional valuation and reports from various points of view, such as the Generally Accepted Accounting Principles (GAAP) international versus local, Sarbanes-Oxley compliance (SOX) and supports the reporting according to International Financial Reporting Standards (IFRS). The solution gives depth of the functionality in accounting, reporting, analyses, financial supply chain and treasury management. Financial and management reporting plus internal controls and documentation make possible a good level of business analytics.

Figure 3.3 describes the various possibilities of the financials.

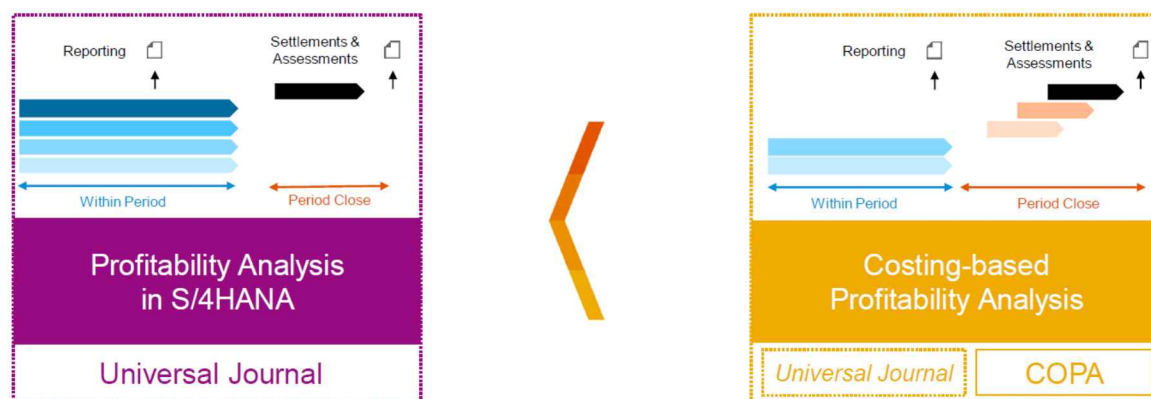
Financial Supply Chain Management	Financial Accounting	Management Accounting	Corporate Governance
• Credit Management • Electronic Bill Presentation and Payment • Collections Management • Dispute Management • In-house Cash • Cash and Liquidity Management • Treasury and Risk Management • Bank Relationship Management	• General Ledger • Accounts Receivable • Accounts Payable • Contract Accounts Receivable and Payable • Fixed Assets Accounting • Bank Accounting • Cash Journal Accounting • Inventory Accounting • Tax Accounting • Accrual Accounting • Local Close • Financial Statements	• Profit Center Accounting • Cost Center and Internal Order Accounting • Project Accounting • Investment Management • Product Cost Accounting • Profitability Accounting • Transfer Pricing	• Audit Information System • Management of Internal Controls • Risk Management • Whistle Blower Complaints • Segregation of Duties

Figure 3.3 Financials

Universal journal

The database for S/4HANA is simplified compared with SAP ERP. This is more than a technical change. The complete integration of Finance and controlling is realized in S/4.

Instant Margin Insight with Profitability Analysis in S/4HANA



Instant margin insight with S/4HANA

Traditional ERP

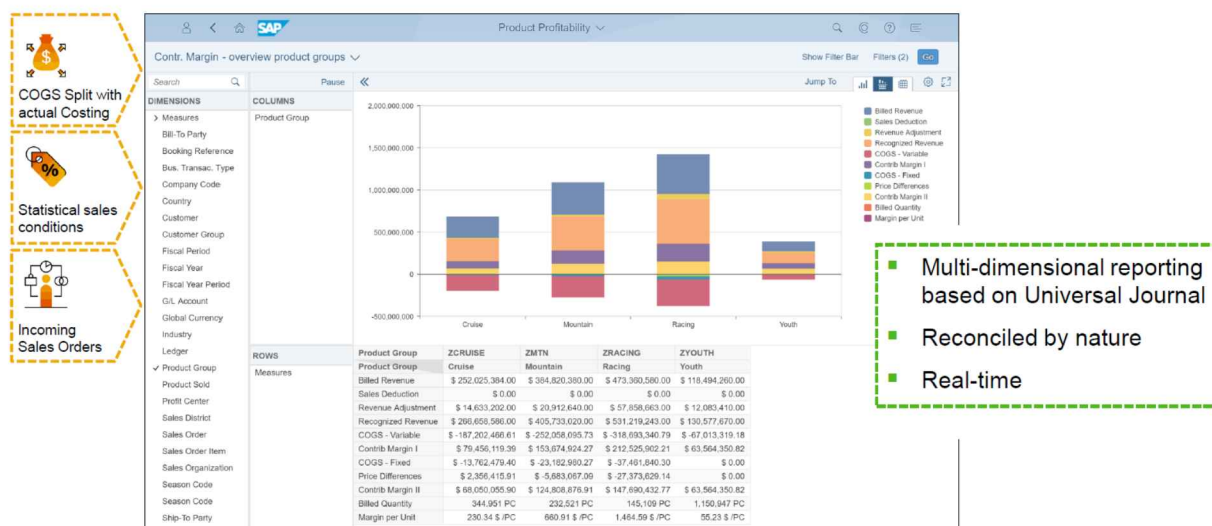
- Separate data model
- Reconciliation challenges
- Not real-time

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For the input of data this means, every GL-account is a Cost Element and all financial postings are posted on a Cost Center. Every financial transaction is recorded in the Universal Journal, this is one table with all fields that might be touched (in ERP we saw separate documents for GL, costing, profit center, etc.).

Reporting is easier, because as a user you can easily create any crosslink analysis between different areas, e.g. cost center postings and profit centers.

Instant Margin Insight with S/4HANA



3.4 HUMAN CAPITAL MANAGEMENT

To meet the demands of today's knowledge-based economy, companies must maximize the potential and productivity of their employees. Maximizing the investment in the organization's human capital is crucial to business success. This requires transforming the HR function from an administration department into a strategic contributor of human capital management strategies. Modern HR departments must deliver a contribution to improve the efficiency and productivity of the employees.

The HCM functions are partly available in S/4HANA and partly in the SAP cloud solution SuccessFactors.

Figure 3.4 describes the possibilities of the various forms of Human Capital Management.

Talent Management	Workforce Process Management	Workforce Deployment
- Recruiting - Career Management - Succession Management - Enterprise Learning - Employee Performance Management - Compensation Management	- Employee Administration - Organizational Management - Global Employee Management - Benefits Management - Time and Attendance - Payroll and Legal Reporting	- Project Resource Planning - Resource and Program Management - Call Center Staffing - Retail Scheduling

Figure 3.4 Human Capital Management

3.5 PROCUREMENT AND LOGISTICS EXECUTION

With SAP S/4HANA Operations, you can manage end-to-end logistics for complete business cycles, such as warehouse management. The procurement will be accomplished in collaboration with the vendor. There are different ways to co-operate with the vendor (collaboration). It is very important to manage the flow of goods very well inside the company (inbound logistics), but also those outside the company (outbound logistics).

Figure 3.5 describes the possibilities of the various forms of the Procurement and Logistics Execution.

Procurement	Supplier Collaboration	Inventory and Warehouse Management	Inbound and Outbound Logistics	Transportation Management
• Requisitioning • Purchase Request Processing • Trading Contract Management • Purchase Order Processing • Receiving • Financial Settlement • Managing Catalog Content • Compliance Management	• Development Collaboration • Purchase Order Collaboration • Confirmation of Goods and Services • Invoice Processing	• Cross Docking • Warehousing and Storage • Physical Inventory	• Inbound Processing • Outbound Processing • Product Classification • Duty Calculation • Customs Communication Service • Trade Document Service • Trade Preference Processing	• Transportation Execution • Freight Costing

Figure 3.5 Procurement and Logistics Execution

3.6 PRODUCT DEVELOPMENT AND MANUFACTURING

With this application it is possible to manage engineering and design, create relevant product data, and plan and execute your manufacturing operations.

This solution enables connectivity to shop floor systems with special quality requirements. With help from Life-Cycle Data Management you can manage production processes, bill of materials, routings, recipes, etc.

Figure 3.6 describes the possibilities of the various forms of Product Development and Manufacturing.

Production Planning	Manufacturing Execution	Product Development	Life-Cycle Data Management
• Production Planning	• Manufacturing Execution • Shop Floor Integration with xMII • Supervision and Control	• Product Development • Development Collaboration	• Document Management • Product Structure Management • Recipe Management • Specification Management • Change and Configuration Management

Figure 3.6 Product Development and Manufacturing

3.7 SALES AND SERVICES

Sales and Service supports a broad scale of activities from presales and sales until after sales. The order processes can be managed. The after sales and service elements provide for good customer relations, through customer relationship management. The complete sales can be managed professionally.

Figure 3.7 describes the possibilities of the various forms of Sales and Service.

Sales Order Management	After Sales and Service	Professional-Service Delivery	Incentive and Commission Management
• Account Processing • Internet Sales • Managing Auctions • Inquiry Processing • Quotation Processing • Trading Contract Management • Sales Order Processing • Mobile Sales • Inbound Telesales • Contract Processing • Billing • Returnable Packaging Management • Consignment	• Phase-In Equipment • Phase-Out Equipment • Asset Scrapping • Product and Warranty Registration • Warranty Claim Processing • Service Contract Processing • Service Plan Processing • Mobile Measurement and Counter Reading • Service Employee Resource Planning • Service Notification Processing • Service Order Processing • Billing • Returns Processing • In-House Repair Processing	• Project Planning and Scoping • Resource and Time Management • Quotation Processing • Sales Order Processing • Project Execution • Time and Attendance • Managing Employee Time and Attendance • Travel Expense Management • Project Accounting • Resource-Related Billing • Resource-Related Intercompany Billing • Billing	• Incentive Business Configuration • Incentive Plan Maintenance • Incentive Processing

Figure 3.7 Sales and Service

3.8 CORPORATE SERVICES

SAP S/4HANA Corporate Services supports and streamlines the administrative processes that are standard to most businesses. Those business-critical support processes must happen with maximum efficiency and control.

The following areas can be managed with Corporate Services:

- Real-estate management.
- Enterprise Asset Management etc.

Figure 3.8 describes the possibilities of the various forms of corporate services.

Real Estate Management	Enterprise Asset Management	Project and Portfolio Management	Travel Management	Environment, Health and Safety	Quality Management	Global Trade Services
• Property acquisition and Disposal • Portfolio Management • Property Management • Technical Management • Support Processes	• Investment Planning & Design • Procurement & Construction • Maintenance & Operations • Decommission & Disposal	• Project Planning • Resource and Time Management • Project Execution • Project Accounting • Prototyping and Ramp-Up • Development Collaboration	• Travel Request and Pre-trip approval • Travel Planning - Online Booking • Travel and Expense Management • Global Travel Policy Compliance • Travel and Expense Analytics	• Product Safety • Hazardous Substance Management • Dangerous Goods Management • Waste Management • Industrial Hygiene and Safety • Occupational Health • Emissions Management	• Quality Engineering • Quality Assurance/Control • Quality Improvement • Audit Management	• Sanctioned Party List Screening • Export Control • Product Classification • Duty Calculation • Customs communication Service • Trade Document Service • Trade Preference Processing • Restitution Handling • Letter of Credit • Periodic Declarations

Figure 3.8 Corporate Services

4 ORGANIZATION DEPICTED IN SAP S/4HANA

There are a number of concepts that form the basic framework of SAP. These concepts, like the corporate structure, are important. All the information within the system is grouped according to this structure. The concepts are explained in this chapter.

The sorts of data in SAP are distinguished first. Next the organizational structure used in SAP S/4HANA is described, and the hierarchy that functions as a backbone in SAP S/4HANA is illustrated. You will use entity relation diagrams (ERD) to depict the relations and links between the different concepts.

4.1 SORTS OF DATA

You can distinguish two sorts of data, namely transaction data and master data.

Transaction data

Transaction data register the daily business transactions. Think in terms of regular sales orders, purchase orders, invoices, production orders and suchlike. SAP gives every document pertaining to a transaction a document number. With transaction data, there are usually a number of related documents, which together register a particular process. A sales process may, for example, involve a quotation, followed by an order, a delivery and a payment.

Master data

Master data are relatively fixed data within a system. In SAP S/4HANA client files, staff files, material files, routings and account schemes are examples of master data. Master data usually have different aspects; these are called views or infotypes. A material always has master data, but storage and accounting data about a particular material can also be stored. You then add views to the material. A member of staff has an address, but also a particular salary and maybe family members. In SAP, these are called infotypes included in employee master data.

Business Partner

Any person or organization that has a relation to the company is called a business partner. The 'General' data, for example name and address, are only related to the partner.

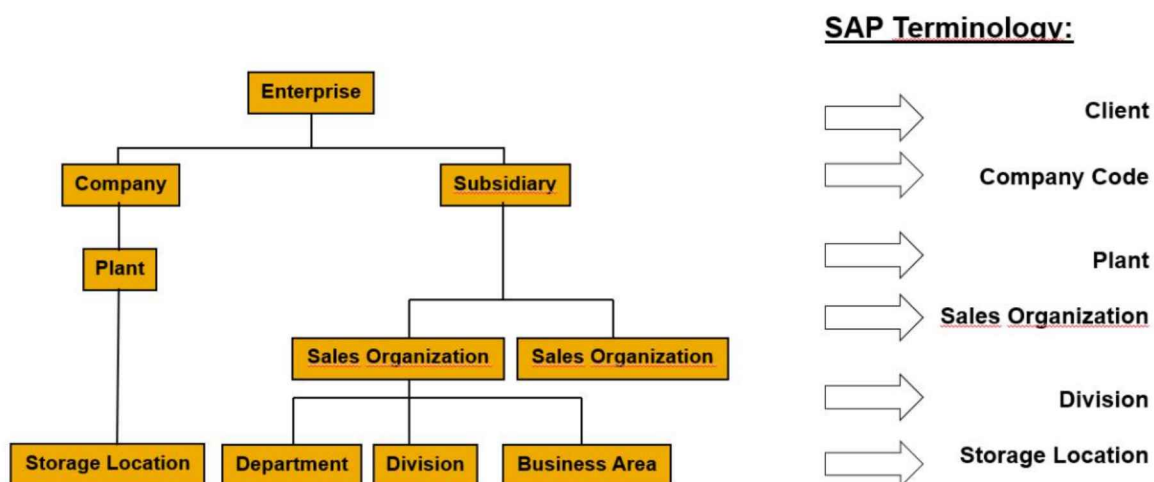
The business partner can be assigned to various roles and those roles can be related to organizational units within the company (see next paragraph). Examples:

Role	Unit
Customer	Sales organization
FI Customer (Debtor)	Company
Vendor	Purchase organization
FI Vendor (Creditor)	Company
Employee	Personnel organization

Several roles can be assigned to one business partner: a company can be both a customer and a vendor; a person can be employee, creditor and customer; etc.

4.2 ORGANIZATION STRUCTURE PROJECTED IN SAP S/4HANA

By defining clients, the administration of several concerns can be conducted within one installation of SAP. With clients, a distinction can be made between the actual business and test or practice businesses that are used to test a new module or to train the work force. By constructing a 'practice client', members of staff can practice with the business data to their hearts' content without influencing the actual business operation.



Client

The client is the highest organizational level (Figure 4.1). There is usually only one client for the entire organization/concern. Several clients could be used for different organizations using the same SAP S/4HANA installation, or for testing and training. Clients have separate and unrelated data.

Company

Each client can contain a number of companies. A company is often used for the subsidiary companies within a concern. Each company is identified as unique by an alphanumeric code of four characters, and is intended as an organizationally and legally independent entity, with its own balance and revenue accountability. SAP makes automatic consolidation possible of the financial accounting for all businesses that have a client. It is also possible to consolidate several clients by using the 'legal consolidation' module.

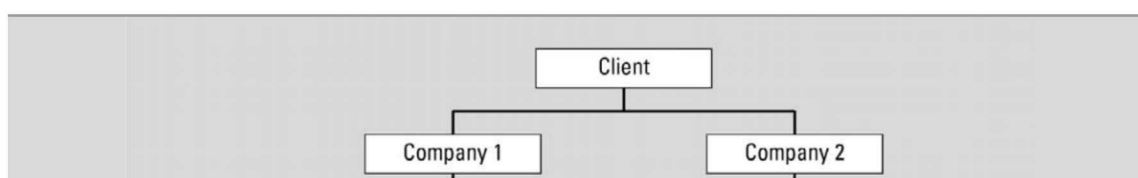


Figure 4.1 Organization structure financial

Controlling area

A controlling area is an organizational unit within a company, used to represent a closed system for cost accounting purposes. A controlling area may include single or multiple company codes that may use different currencies. These company codes must use the same operative chart of accounts. All internal allocations refer exclusively to objects in the same controlling area.

Operating concern

Operating concerns are the accountability areas within a business that are used for the allocation of costs to cost centers. Each cost center is allocated to an operating concern. Each operating concern can represent more than one company.

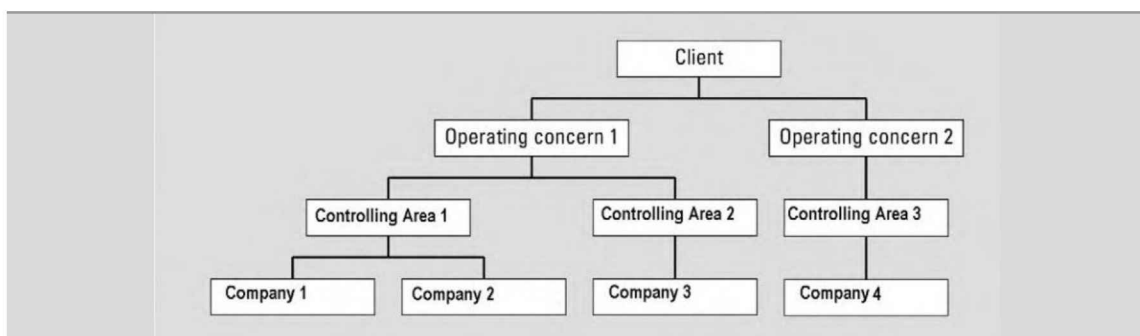


Figure 4.2 Organization structure controlling

Plant

Businesses can also be subdivided into operational facilities called plants, which are used in the logistic module. Various storage locations can be allocated to each plant. Plants and business areas are not mutually exclusive. For example, a plant can make various products, which are paid for in different business areas. In the same way, a business area can handle products from various plants.

Storage location

The organizational unit storage location can consist of various storage areas. Storage locations allow the differentiation between the various stocks of a material in a plant. The units are important for the valuation (FI) and planning (PP).

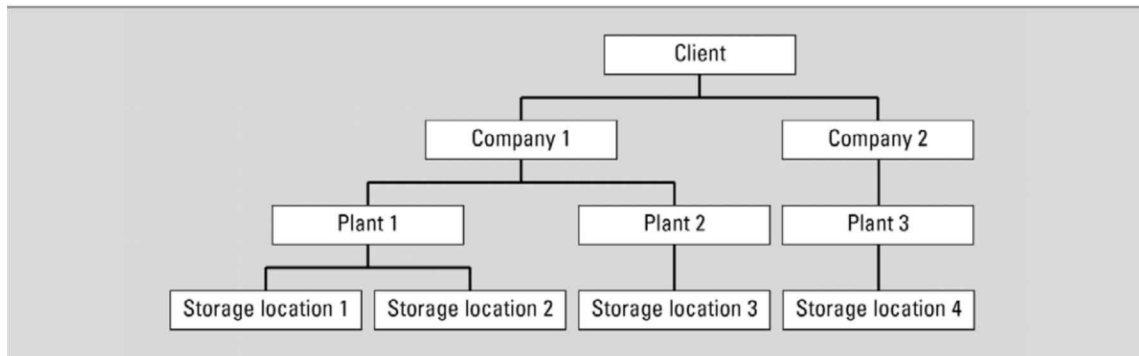


Figure 4.3 Organization structure logistics

4.3 PROJECTING THE BUSINESS STRUCTURE IN SAP S/4HANA

Usually, there is more than one way of depicting the concepts/departments as they exist in the enterprise. The different possible representations will, however, have certain consequences. If a particular alternative is simpler to implement, the result of this will frequently be that certain information needed in the longer term cannot be obtained from the system. Conversely, a more complex alternative will have the consequence that more information can be obtained from the system, but the users will need more training to use the system properly and to interpret the information correctly.

Example 4.1

A university has decided that each faculty be depicted in SAP as a company with a company code, instead of having one company code for the whole university.

Once a particular set-up has been chosen, it may be difficult to change it. Due consideration should therefore be given to this set-up: it is important not only to look at current information needs, but also the information that the system may be required to process in the future.

4.4 SAP HIERARCHY

Within SAP, there are a number of concepts which can be considered as the framework of the system. They form the so-called 'corporate structure' or organizational structure. These concepts are important because all information that is registered in the system is linked to these concepts. Many of these concepts will be explained in this chapter. In the first place, the SAP hierarchy is explained; this can be considered to serve as the backbone for the concepts. Then the concepts of the most important modules will be illustrated. In doing this you will use ERDs (Entity-Relation-Diagrams) to clarify the relation /allocations between the separate concepts. In the ERD there is an indication of which concepts are related to each other and whether there are any 'n-on-m' relations.

Client

The highest level distinguished is that of the client. On logging in, this must be instantiated. Each client is a detached, individual work environment. Frequently, within one enterprise use is made of a production environment (the actual system), a training environment and a test environment.

Nearly all customizing activities are related to one client. In other words, an activity only affects the client in which a user is working.

Company code

The most important concept within a client is the company code. A company code represents an independent legal entity with fiscal obligations. All data relating to the flow of funds are linked to the company code (see Figure 4.4). All separate financial concepts should ultimately lead to one balance sheet and one profit and loss statement, which is sent to the tax authorities (and the shareholders).

The most important data that are recorded are the following:

1. Chart of Accounts (general ledger accounting)
2. Currency
3. Address

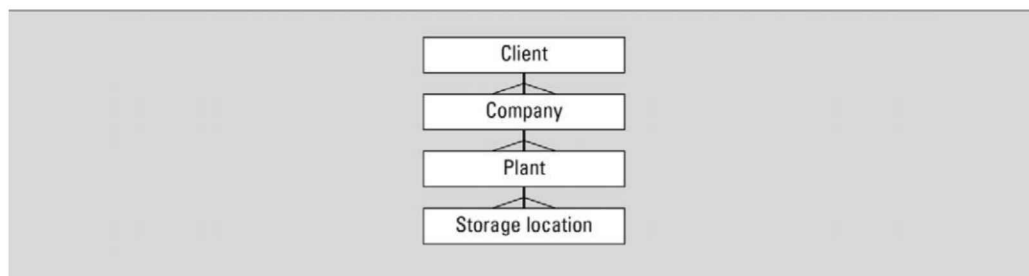


Figure 4.4 SAP hierarchy

Plant

The plant is the concept to which logistical data are linked. As Figure 4.4 shows, there is an allocation from the plant to the company code. This means that the financial consequences of a logistical process can be passed on directly (provided they have been implemented) to the financial modules.

In setting up a plant, only the address must be entered. At a later stage, all sorts of control parameters will be linked to the plant. Control data are data that determine how the system reacts in different situations.

Example 4.2

Various sorts of materials will be entered in various general ledger accounts in the accounts scheme. This takes place by means of control parameters.

Storage location

Storage locations are used for the stock registration of materials. With a storage location, only a code and a specification are recorded. The reason that no address is recorded is that it is assumed that the storage location is situated in the same place as the plant.

Storage locations are important in SAP for very complicated stock registrations, production planning (PP) and requirement planning. The modules concerned will not be discussed in this chapter.

4.5 FINANCIAL (FI)

The organization structures in Financial can be subdivided into:

- Group company
- Controlling area
- Profit centre

Group company

If a company has a number of subsidiary companies, it is also obliged to issue consolidated annual accounts. By way of the group company SAP provides the possibility of having this consolidation executed entirely by the system.

Naturally, adjustments will have to be made here too (for example, to exclude reciprocal deliveries between parent company and subsidiary etc.). These adjustments are linked to the group company.

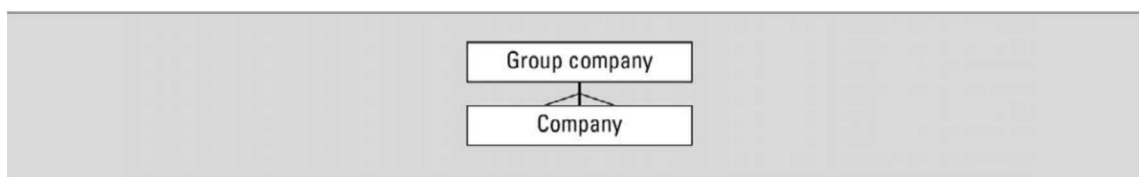


Figure 4.5 Group company

Controlling area

An organizational unit within a company, used to represent a closed system for cost accounting purposes. A controlling area may include single or multiple company codes that may use different currencies. These company codes must use the same operative chart of accounts. All internal allocations refer exclusively to objects in the same controlling area.

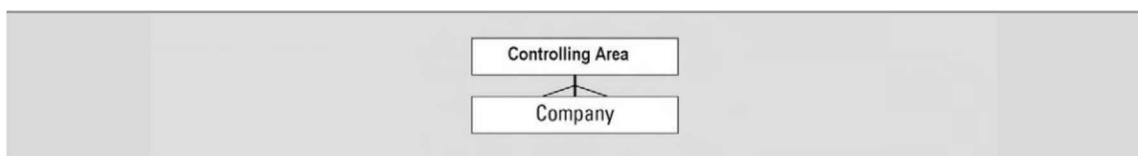


Figure 4.6 Controlling area

Example 4.3

A international enterprise has several subsidiaries in the Netherlands. Cost can be analysed per country with the use of a controlling area for the Netherlands.

Profit center

As can be seen from Figure 4.7, the profit center is an unrelated concept, not linked to any other concept. A profit center is used to produce balances and Profit and Loss accounts. The profit center is not an independent legal entity. It does not therefore have to comply with the legal requirements for external accounting.

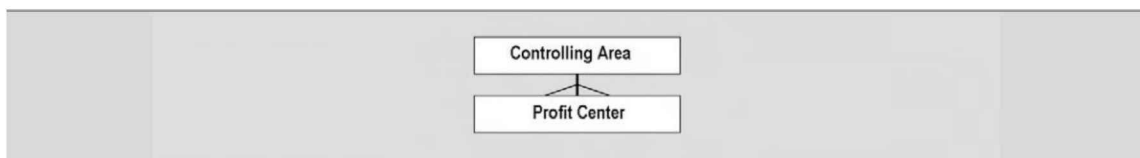


Figure 4.7 Profit center

In a company, it is often standard practice to assign some profit accountability to departments. This profit is generated by the services/products that they deliver to other departments or customers. By using profit centers this can be registered in SAP.

Example 4.4

An international enterprise can analyze the costs and revenues per product group by creating within the controlling area the next profit centers: Printers, Desktops and Laptops.

4.6 MATERIAL MANAGEMENT (MM)

Purchasing organization

All data relating to suppliers, delivery addresses and conditions of payment are allocated to the purchasing organization. Figure 4.8 shows that there is an allocation both with the financial concepts and with the logistical concepts. In the case of procurement and suppliers, both financial and logistical information is recorded. Of course, a purchasing organization can only be allocated one company code. The conditions of payment and receipt of goods are accordingly entered in one account scheme. A company may have several purchasing organizations. The subdivision into several purchasing organizations may, for example, be based on location or on the sort of products that are being purchased.

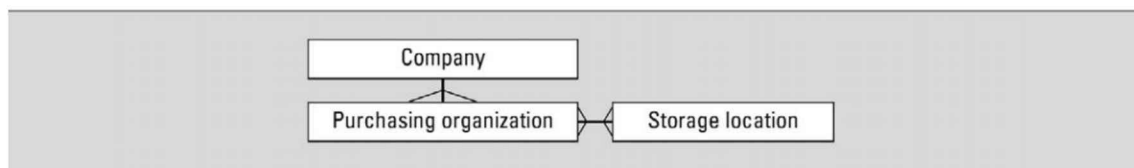


Figure 4.8 Purchasing organization

Furthermore, it is possible to make purchases for several plants. It can also occur that a plant can be served by several purchasing organizations. Although it is not indicated in the ERD in Figure 4.8 that there is a relationship between the plant and the company code, nevertheless, it does exist. The fact is that the purchasing organization can only be linked to the plants that fall under the same company code. If that were not the case, the system would not know under which company code which part of the profit should be entered.

Example 4.5

A wholesaler supplies car components (batteries, towbars and tires) to repair companies. It has three distribution centers for this in Amsterdam, Rotterdam and Eindhoven, where the purchasing department is located. Its activities are subdivided into batteries, towbars and tires. In SAP, this is realized by using three plants (= three logistical streams), one for each distribution center. In addition, three purchasing organizations have been set up which are subdivided into the various product groups. Each purchasing organization is linked to each plant, so that each distribution center can be furnished with the various product groups.

4.7 SALES AND DISTRIBUTION (SD)

The organization structures in Sales and Distribution can be subdivided into three groups:

- Externally oriented sales organization
- Internally oriented sales organization
- Distribution

4.7.1 EXTERNALLY ORIENTED SALES ORGANIZATION

The concepts concerned here describe the market and the possibilities which are offered to customers relating to deliveries, price conditions etc. This includes the following four entities:

- Sales organization
- Distribution channel
- Product area
- Sales area

Sales organization

This is an organization unit that is responsible for the sale of particular products or services. This sales unit also has legal liability in relation to damages. A complete sales transaction takes place within one sales organization. (This sales translation consists of quotation P order P delivery P invoice). Often, sales organizations mirror the division of a particular market geographically.

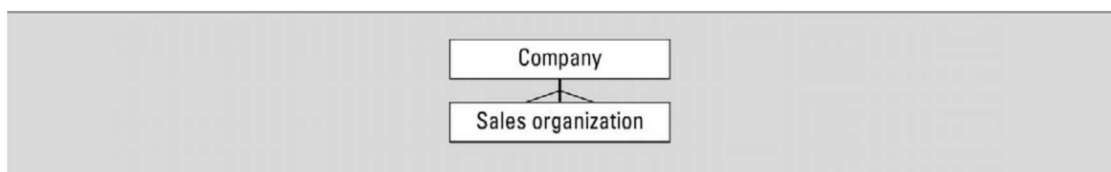


Figure 4.9 Sales organization

Example 4.6

The wholesaler mentioned before supplies his products to car repair companies located all over the Netherlands. For this purpose, there are three sales organizations operating in the northern, central and southern Netherlands, respectively.

Distribution channels

A distribution channel denotes the manner in which products or services arrive at the customer. A more usual term for distribution channel is sales route. As the diagram shows, one distribution channel can be used in several sales organizations. A sales organization can make use of several distribution channels.

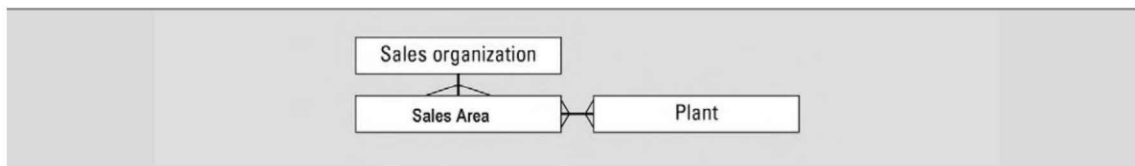


Figure 4.10 Distribution channel

Example 4.7

The wholesaler makes use of three different distribution channels:

- Internet
- Franchise chains
- Other garages

All three sales organizations use the channel franchise chains and other garages. Sales organization central Netherlands also delivers via the Internet.

A Distribution channel also has an allocation with a Plant. Each sales organization is free to determine Distribution channels, through which the plants can be supplied. For the combination of Distribution channel and sales organization it can therefore be established from which plant (or plants) supplies can be provided. A Distribution channel can get supplies from several plants, and a plant can have several Distribution channels allocated to it.

Example 4.8

Central Netherlands supplies by Internet from Rotterdam or Eindhoven. The necessary assignments are:

- Internet – Central Netherlands – Rotterdam
- Internet – Central Netherlands – Eindhoven

Division

A division is used to group materials, products or services. A material is grouped in one division. Price agreements with a particular client can be allocated to these groupings. In addition, sales statistics can be generated per division. Divisions can be used in several sales organizations, while within one sales organization products from several divisions can be sold.

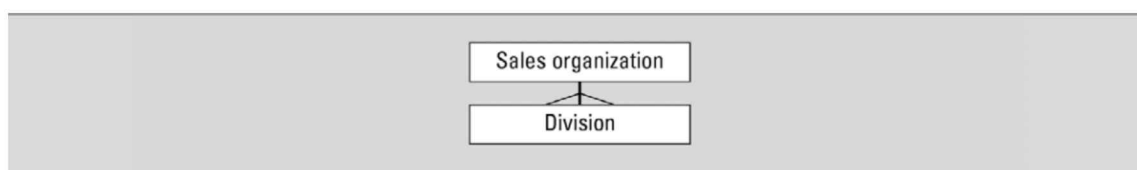


Figure 4.11 Division

Example 4.9

The products are divided into three divisions: Batteries, Tow bars and Tires (these are the same divisions as in the Purchasing department). Tires are sold in each sales organization. Batteries are not sold in Southern Netherlands, tow bars are not sold in Northern Netherlands.

Sales area

The sales area diverges from the three concepts named above. In fact, in the case of a sales organization, distribution channel and a division, a code and a description are established. With a sales area this is not established. A sales area is understood to be the combination of one sales organization, one Distribution channel and one division. This consists therefore of one allocation. With the sales area it is established through which distribution channels a group of materials (division) can be delivered. In view of the fact that this can vary per sales organization, it is established per sales organization.

The distribution channels and divisions are already allocated to the sales organization. The allocation of the distribution channel and the division within the sales organization can thus only be effected with the distribution channels and the divisions that are allocated to the sales organization concerned.

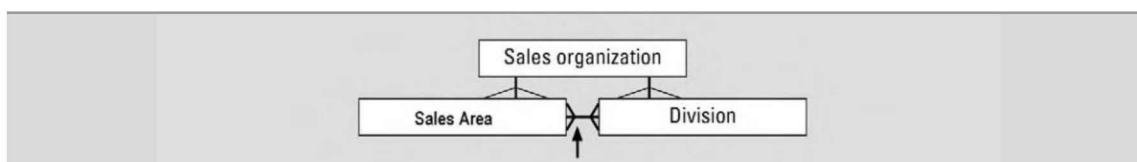


Figure 4.12 Sales area

Example 4.10

For the Central Netherlands sales organization you have the following distribution channels (see the examples above): Internet, Franchise chains and other garages. Furthermore, there are the following divisions: Batteries, Tow bars, Tires.

The following sales areas have been assigned:

- Central Netherlands – Internet – Batteries
- Central Netherlands – Internet – Tires
- Central Netherlands – Franchise chains – Batteries
- Central Netherlands – Franchise chains – Tow bars
- Central Netherlands – Franchise chains – Tires
- Etc.

4.7.2 INTERNALLY ORIENTED SALES ORGANIZATION

These concepts describe how the sales department can be organized within the company (with respect to accountability etc.):

- Sales office
- Sales group

Sales office

The sales office is the department of the company that is responsible for the sale of particular products through a particular distribution channel.

As Figure 4.13 shows, several sales areas can be allocated to a sales office. A sales area can be used by several sales offices. As the sales office is allocated to the sales area, it is therefore also allocated to the sales organization. The sales organization is in its turn allocated to the company code, so that the results can be automatically entered into the financial modules.

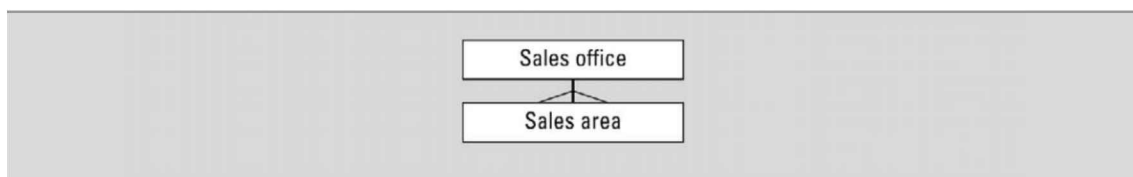


Figure 4.13 Sales office

Example 4.11

In the northern Netherlands there are two sales offices – one for sales through the route run and one for the express run. It would also have been possible to subdivide the sales offices into the various product groups or sorts of clients.

Sales group

This category makes it possible to group together the sales assistants who are involved in the same sales activities. A sales group can perform sales activities for several sales offices. Several sales groups can be operative in one sales office. These sales groups make it possible for various accountability areas to be defined within the sales office.

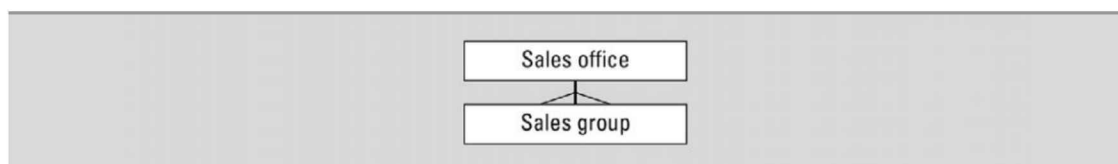


Figure 4.14 Sales group

Example 4.12

Northern Netherlands has three sales groups: route run, express run and representatives. The route run and the express run groups receive all the orders. The representatives visit the most important clients to maintain contacts and to take orders. They can take orders for both the route run and the express run and they are therefore assigned to both sales offices. The route run and express run are only assigned to one sales office (route run office and express run office respectively).

4.7.3 DISTRIBUTION

This paragraph describes basic elements of shipment processing organization. This includes:

- Shipping point
- Loading point

Shipping point

From the shipping point the goods are sent to the customer. This location can be physical (e.g. a collection of loading bays at the storage location), but it is also possible to define a group of employees as a shipping point (a group of employees who are responsible for particular shipping activities). A shipping point can service several plants, while plants can also make use of several shipping points.

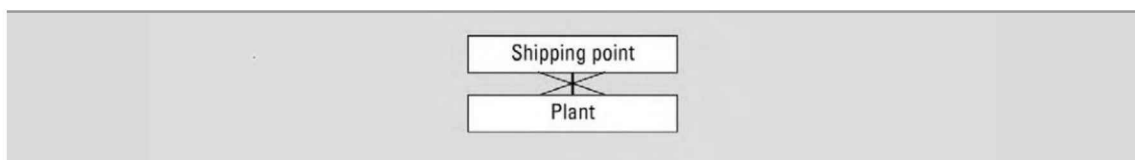


Figure 4.15 Shipping point

Example 4.13

As mentioned earlier, the wholesaler has set up three plants (=three logistical flows) in SAP, with a corresponding storage location. Three shipping points have also been set up – Amsterdam, Rotterdam and Eindhoven.

Loading points

The SD module can also be used if no loading points have been established. If use is to be made of the extensive possibilities relating to stock registration, delivery loading points should be established. As Figure 4.16 shows, a loading point can only be allocated to one shipping point.

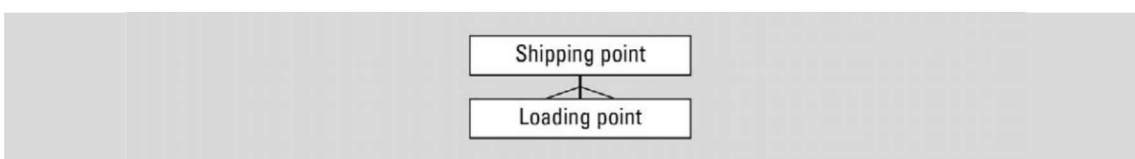


Figure 4.16 Loading point